Section 1 — Logic & Problem Solving

Soal 1 — Marble Problem

Jika terdapat:

- 8 kelereng
- 7 kelereng berbobot 100 gram
- 1 kelereng berbobot 125 gram

Bagaimana cara menemukan kelereng terberat menggunakan timbangan neraca dua lengan maksimal 2 kali penimbangan?

Jelaskan:

- langkah algoritma,
- reasoning,
- dan complexity thinking.

Soal 2 — System Design Thinking

Sebuah platform e-commerce memiliki:

- 10.000 transaksi per hari,
- proses pembayaran,
- email invoice,
- notifikasi WhatsApp,
- dan dashboard admin realtime.

Jelaskan:

- bagaimana arsitektur backend yang akan dibuat,
- bagaimana scaling dilakukan,
- bagaimana mencegah duplicate event,
- bagaimana logging dan monitoring diterapkan,
- serta kapan harus menggunakan queue/message broker.

Bebas menggunakan diagram.

Section 2 — Backend API Development

Requirement

Buatlah REST API menggunakan salah satu:

- Node.js (Express/NestJS/Fastify)
- Golang
- Laravel
- .NET
- Spring Boot

Fitur Wajib

Authentication

- Register
- Login
- JWT Authentication
- Role based access

Product API

CRUD Product:

- Create Product
- Update Product
- Delete Product
- Get Detail Product
- Product Pagination
- Search Product

Order API

- Create Order
- Order Detail
- Order History

Validation

- Request validation
- Error handling standardization

Section 3 — Database & Optimization

Peserta wajib:

- menggunakan relational database,
- membuat migration,
- membuat indexing yang relevan,
- menjelaskan alasan struktur database.

Tambahan:

- soft delete,
- transaction handling,
- query optimization.

Section 4 — Async Processing & Queue

Buat asynchronous service menggunakan:

- RabbitMQ
- Kafka
- Redis Queue
- atau sejenisnya.

Studi Kasus

Saat order berhasil:

- kirim email invoice,
- simpan activity log,
- kirim notifikasi async.

Requirements:

- retry mechanism,
- dead letter handling,
- idempotency handling.

Section 5 — Testing

Minimal testing coverage:

- Unit Test
- Integration Test

Gunakan:

- Mocha/Jest
- PHPUnit
- Go test
- xUnit
- atau framework sesuai stack

Coverage minimal:

- 70%

Peserta wajib menjelaskan:

- strategy testing,
- mocking,
- dan edge cases.

Section 6 — DevOps & Deployment

Peserta wajib:

- menggunakan Docker,
- membuat docker-compose,
- environment separation,
- health check endpoint.

Nilai tambahan:

- CI/CD
- Kubernetes
- reverse proxy nginx
- HTTPS setup
- monitoring stack

Section 7 — Observability

Tambahkan:

- structured logging,
- request tracing,
- metrics sederhana.

Contoh:

- Winston / Pino
- OpenTelemetry
- Prometheus

Section 8 — Security

Implementasikan:

- rate limiting,
- password hashing,
- SQL injection prevention,
- CORS handling,
- secure headers,
- environment secret management.

Section 9 — Documentation

Repository harus memiliki:

- setup guide,
- architecture explanation,
- API documentation,
- ERD/database schema,
- flow async processing.